

Domestic violence charge records and ambient heat

Revised methods and restricted-sample sensitivity analysis

September 24, 2026 • Recorded-data period: July 2018–December 2025 • Model period: 2021–2024

Structured abstract

Objective. Reassess the association between modeled temperature and daily counts of source identifiers carrying qualifying domestic-violence charges after correcting weather-day boundaries and evaluating specified sources of measurement and model uncertainty.

Population and estimand. Three supplied exports contain 12,478 core-DV identifiers and 15,161 expanded-DV identifiers. These administrative units are not verified distinct people, arrests, offenses or convictions. Models estimate conditional daily count ratios, not population incidence or causal effects. [1,21]

Design. The preserved full-sample reference includes 6,637 eligible DV identifiers. Provider quota limited complete revised hourly weather to 64 of 112 ZIP areas, containing 6,501 eligible DV identifiers (98.0% of the original sample). The old and corrected weather were therefore compared on the same restricted sample. All 11 DV revision specifications and a joint 24-test correction spanning DV and general-record outcomes are reported. This is a retrospective amendment, not a preregistered or independent replication. [21]

Methods. Hourly weather was grouped into America/Los_Angeles civil dates. Poisson log-link models retained ZIP, year-month and weekday effects; sensitivity analyses addressed warrant co-charges, humidity, precipitation, holidays, local seasonality, nonlinear exposure, alternate NOAA observations and daylight-saving day length. Daily summed-score HAC 7 uncertainty was used. [9,10,15,22,23,30,31]

Results. The original full-sample estimate was +2.89% (95% CI, –1.05% to +6.98%) per 10°F. On the restricted sample, old-weather and civil-day count ratios were 1.027 (0.988–1.069) and 1.027 (0.988–1.069), respectively. For the civil-day reference, raw $p=0.179$ and joint Holm $p=1.000$. No DV revision test met the 0.05 threshold after the joint Holm correction.

Interpretation. The civil-day reference interval includes no association. The incomplete geographic correction does not replace the full original analysis. Remaining uncertainty includes source completeness, ambiguous recorded-date/location meanings, selective DV-charge identification, modeled exposure and unmeasured changes in reporting or enforcement. Removing known warrant co-charges does not verify offense timing.

1. Administrative sources and coverage

The requester supplied three CSV exports: 214,745 raw lines, of which 214,741 valid charge rows form 136,313 source-scoped IDs; four lines lack usable identifiers/dates. Original files and hashes are preserved. The available data do not establish a countywide census of all municipal-police arrests. Every 2024 and 2025 row labels Agency as SHERIFF; that column is absent from the older export. Agency labels and a regional system name do not substitute for an agency response letter or extraction specification. [1,21,29]

Source export	ID header	Date header
July 2018–2023	Incident Number	Incident Date_Time (trailing space in raw header)
2024	Incident Number	Arrest Date/Time
2025	Arrest Ref Nbr	Arrest Date/Time

The 2024 identifier is Incident Number; the identifier header changes only in 2025. The equivalence of the two date fields, their time zone, and their relationship to offense occurrence remain unverified. City and ZIP also lack a verified role such as offense, arrest, residence or booking location.

Recorded year	All source IDs	Core DV	Expanded DV
2018	12,092	988	1,215
2019	24,760	1,947	2,331
2020	18,589	1,838	2,254
2021	19,018	1,875	2,351
2022	18,317	1,674	2,104
2023	18,096	1,682	2,019
2024	17,121	1,574	1,856
2025	8,283	900	1,029
Conflicting year	37	0	2

These unchanged descriptive totals precede model eligibility screens. 2018 begins July 1. The 2025 file has 79 dates with no raw record and 83 dates without an unambiguous-date grouped ID; these are distinct measures. January–May 2025 contains only 213 charge rows and 144 monthly ID appearances. No 2026 export was supplied. The 2025 discontinuity is retained as a coverage limitation, not interpreted as a crime decrease.

2. Outcome definition and identifier grouping

The core outcome counts one source-file-scoped identifier if any associated charge matches the reviewed exact code/type/description map. The mapping covers 1,603 observed combinations. It identifies selected relationship-specific charges; it does not recover every form of domestic or intimate-partner violence. [1,2–8,20,21]

Tier	Qualifying charges	Interpretation
Core DV	PC 243(e)(1); 273.5(a)/(f)(1)	Relationship-specific battery or corporal injury. [2,3]
Core DV	Historical PC 262 variants; 12022.7(e)	Recorded spousal-rape charges or DV injury enhancement. Repeal of 262 limits consistent identification after 2021. [4,5,8]
Expanded only	PC 273.6(a)/(b); 166(c)(4), with domestic-labeled descriptions	Protective-order statutes are broader; a domestic relationship is not independently verified. [6,7]
Unconfirmed	General assault, threats, stalking, child/elder abuse, other court-order charges	Not classified DV unless a qualifying charge occurs on the same source ID. Absence of a qualifying charge does not prove absence of DV.

Core DV comprises 12,487 charge rows and 12,478 source IDs; the broader union adds 2,683 order-only IDs. Source scoping avoids collisions between exports. Grouping avoids counting each charge as a separate event, but one source ID may still cover multiple people or episodes. There are 926 identical charge rows; retaining them in source files does not inflate distinct-ID totals.

Metadata conflicts and selection

Across all groups, 37 conflict on year, 121 on month, 172 on date, 419 on ZIP and 565 on full timestamp. Conflicting date/ZIP assignments are not resolved by selecting a first value. Core-count stability under composite-key checks is a reproducibility result, not a validation that each group is one arrest. Case-level DV flags, victim relationships, narratives and dispositions were absent. [21]

Warrant co-charge screen

The baseline excludes warrant subtypes and specified detention/court classifications but can retain ADULT groups with warrant charges. The expanded screen uses exact PC 978.5, 827.1, 1551(a) and the reviewed ZZ warrant aliases on any associated line. It removes 67 of 6,637 full-sample DV groups; in the revised geographic subset it removes 66 of 6,501. R4–R8 use the stricter outcome. PC 1551.1 explicitly permits arrest without a warrant and is not flagged by that code alone. [21,32,33]

3. Weather reconstruction and restricted coverage

Exposure remains modeled ambient weather at Census ZIP Code Tabulation Area representative points. It is not temperature measured where a person or alleged offense was located. The original 112 postal areas shared 50 ERA5-Land grid cells; ZIP fixed effects do not create independent local weather measurements. Representative coordinates and the original returned elevations were retained for the revision. [9–11,25]

Local civil dates

The released daily data used a fixed UTC–7 window. The revision requests hourly UTC timestamps and converts them using historical America/Los_Angeles time-zone rules. Temperature maximum/minimum and arithmetic mean, and relative-humidity mean, use timestamps from local midnight up to the next midnight. Civil dates can contain 23, 24 or 25 hours. Hourly precipitation accumulations use their interval-ending timestamps, with sums covering the same local interval. [9,14,30]

Primary civil-day duration	Calendar dates
23 hours	4
24 hours	1453
25 hours	4

Comparing start and end boundaries, 513 of 1,461 primary dates have at least one boundary different from the fixed UTC–7 window. This boundary count is distinct from the number of cells whose temperature value changes. A repeated nighttime hour can change the interval even when midnight has the daylight-saving offset.

Source and retrieval differences

Temperature and humidity come from the specified ERA5-Land hourly source; precipitation is obtained separately from coarser ERA5 hourly data because the ERA5-Land API did not supply it. The revision separates civil-minus-new-fixed-window differences from new-fixed-minus-released differences. Newly retrieved fixed-window maxima and minima exactly match the released values on this subset; daily mean temperatures also differ because of aggregation and rounding. The table quantifies the daily-high correction. [9,10,21,25,31]

Daily maximum comparison	Mean absolute difference, °F	Maximum absolute difference, °F
Civil minus released	0.000499	1.500
Civil minus new fixed window	0.000499	1.500
New fixed window minus released	0.000000	0.000

Restricted availability. Complete revised weather was acquired for 64 of 112 ZIP areas before the provider quota blocked further acquisition. Requests prioritized ZIPs with more source rows; this is not a random geographic sample. The retained areas contain 98.0% of originally eligible DV IDs, which does not establish representative geographic coverage. Missing areas are not filled with old weather. [21]

4. Statistical estimand and model

The response is the number of eligible core-DV source identifiers in ZIP z on recorded civil date d . A full calendar is constructed within each retained ZIP, including zero-count dates. R0 estimates the ratio of conditional expected daily counts for a 10°F higher modeled maximum, under a common log-linear association. It is not a per-person crime rate. [16,21]

$$\log \mu_{zd} = \alpha_z + \gamma_{m(d)} + \delta_{w(d)} + \beta \frac{T_{zd} - 70}{10}$$

$$\mu_{zd} = E[Y_{zd} | x_{zd}]$$

Y_{zd} is the daily count; T_{zd} is maximum temperature in °F; x_{zd} denotes model covariates. α_z includes the intercept and ZIP effect; $\gamma_{m(d)}$ and $\delta_{w(d)}$ are year-month and weekday effects. The coefficient β corresponds to a 10°F increase; centering at 70°F changes the intercept, not that contrast.

Reference sample	DV identifiers	Fitted ZIPs	Fitted ZIP-days
Original full sample	6,637	95	138,795
R0/R0 paired geographic subset	6,501	63	92,043
R4: stricter warrant subset	6,435	63	92,043
R5: combined model	6,435	63	89,125

One of the 64 available ZIPs has no core-DV outcome during the period, leaving 63 fitted ZIPs in R0. All-zero ZIPs lack a finite count-model intercept and are omitted from estimation; their zero cells remain in the calendar/audit. Station-paired models can have missing observation dates; all 1,461 calendar dates remain in the score sequence.

Covariance and intervals

Model scores are summed across ZIPs within each date. Bartlett-weighted Newey-West covariance uses seven calendar-day lags and the $N/(N-K)$ covariance correction, with N fitted ZIP-days and K parameters. This permits cross-ZIP score dependence and short-lag temporal dependence; it does not treat each ZIP-day as an independent replicate. Exponentiated normal Wald intervals are individual 95% intervals. HAC 14 / HAC 28 checks assess lag sensitivity. [15,22,23]

Offsets and interpretation

R0 estimates counts per civil date. R9 adds the offset $\log(h_{zd}/24)$, where h_{zd} is elapsed hours, imposing proportional exposure opportunity on 23/25-hour dates. It is a sensitivity because the lost/repeated hour is nighttime and record intensity need not be uniform. A fixed ZIP population offset is absorbed by unrestricted ZIP intercepts; it cannot address changing visitors or agency coverage. No compatible time-varying population denominator was supplied.

5. Retrospective revision specifications

The methodological amendment was frozen before revised estimates were computed, after the original results and external feedback were known. A provider-quota amendment added the matched old-weather fit and restricted geography before these estimates. This is retrospective, feedback-driven analysis, not preregistration. All specified fits are reported; no model is chosen by its result. [21]

ID	Specification	Definition
R00	Old-weather paired reference	Original released daily maximum on exactly the revised available ZIP-days.
R0	Civil-day reference	Corrected civil-day maximum; original eligibility retained within available ZIPs.
R1	Previous-day maximum	Prior civil-date maximum replaces contemporaneous maximum.
R2	Additional weather/calendar controls	R0 plus observed federal holiday, $\log(1+\text{precipitation in mm})$, and mean relative humidity per 10 percentage points.
R3	Local annual seasonality	R0 plus separate annual sine/cosine terms for each ZIP; leap-year phase handled explicitly.
R4	Warrant co-charge exclusion	R0 with any reviewed warrant co-charge excluded.
R5	Combined adjustment	R2+R3+R4.
R6	Nonlinear combined model	R5 with a restricted cubic spline; four weather-only knots fixed across outcomes.
R7	Paired station sample, modeled exposure	R5 on the NOAA-eligible ZIP-day subset, retaining ERA5-Land maximum.
R8	Paired station sample, observed exposure	Exactly R7 rows/counts, replacing maximum with NOAA station maximum.
R9	Civil-day duration offset	R0 plus $\log(\text{elapsed hours}/24)$.

All fits include ZIP, year-month and weekday effects. Observed federal holidays are a proxy for selected calendar disruptions, not a complete set of local events or school schedules. Humidity/precipitation controls and ZIP-specific seasonality do not establish a valid causal adjustment set.

Multiplicity

The revision family contains 24 tests across two outcomes: 10 linear temperature coefficients per outcome (R00, R0–R5, R7–R9), one joint spline association and one spline nonlinearity test per outcome. Holm correction is applied jointly, not separately within DV. Confidence intervals and the plotted spline band remain nominal, not familywise. Original testing families are archived separately; the new correction does not erase earlier analysis choices. [17,24]

6. Domestic-violence temperature estimates

The original full-sample result and the paired comparisons answer different questions. R00 versus R0 changes weather measurement while holding the restricted records and fitted rows fixed. Comparing the original full sample with R00 also changes geographic coverage. Neither comparison identifies a causal heat effect.

Reference comparison	Count ratio (95% CI)	DV IDs
Original full sample	1.02888 (0.98952–1.06981)	6,637
R00: old weather, restricted sample	1.02736 (0.98762–1.06869)	6,501
R0: civil-day weather, same sample	1.02741 (0.98768–1.06873)	6,501

Specification	Count ratio (95% CI)	Raw <i>p</i>	Holm <i>p</i>	DV IDs
R00	1.027 (0.988–1.069)	0.180	1.000	6,501
R0	1.027 (0.988–1.069)	0.179	1.000	6,501
R1	1.041 (1.001–1.083)	0.044	0.964	6,501
R2	1.040 (0.983–1.101)	0.170	1.000	6,501
R3	1.031 (0.990–1.075)	0.144	1.000	6,501
R4	1.025 (0.986–1.065)	0.218	1.000	6,435
R5	1.048 (0.988–1.112)	0.116	1.000	6,435
R6: joint association	Spline; no single ratio	0.181	1.000	6,435
R6: nonlinearity	Spline; no single ratio	0.313	1.000	6,435
R7	1.065 (0.979–1.159)	0.142	1.000	3,809
R8	0.979 (0.912–1.051)	0.557	1.000	3,809
R9	1.027 (0.988–1.069)	0.179	1.000	6,501

Linear ratios are per 10°F. Specification definitions appear on the preceding page. R6 has three temperature-basis parameters; its joint test and curvature test are reported separately. Holm values use all 24 revision tests. These confidence intervals are not multiplicity-adjusted.

The civil-day reference interval includes no association. No DV revision test met the 0.05 threshold after the joint Holm correction. The conditional R0 estimate is +2.74% (95% CI, –1.23% to +6.87%) per 10°F. The combined R5 estimate is +4.85% (95% CI, –1.16% to +11.22%). These are sensitivity comparisons, not a search for a preferred significant coefficient.

7. Nonlinear temperature and model diagnostics

R6 uses the combined controls and strict warrant outcome. Four knots are fixed at the pooled weather-only 5th, 35th, 65th and 95th percentiles within the available primary panel, shared across both outcomes. The basis has three temperature coefficients and linear tails. This avoids selecting knots by outcome significance, but it remains a post hoc design. [21]

Knot temperatures (°F): 55.50, 66.10, 75.10, 89.80.

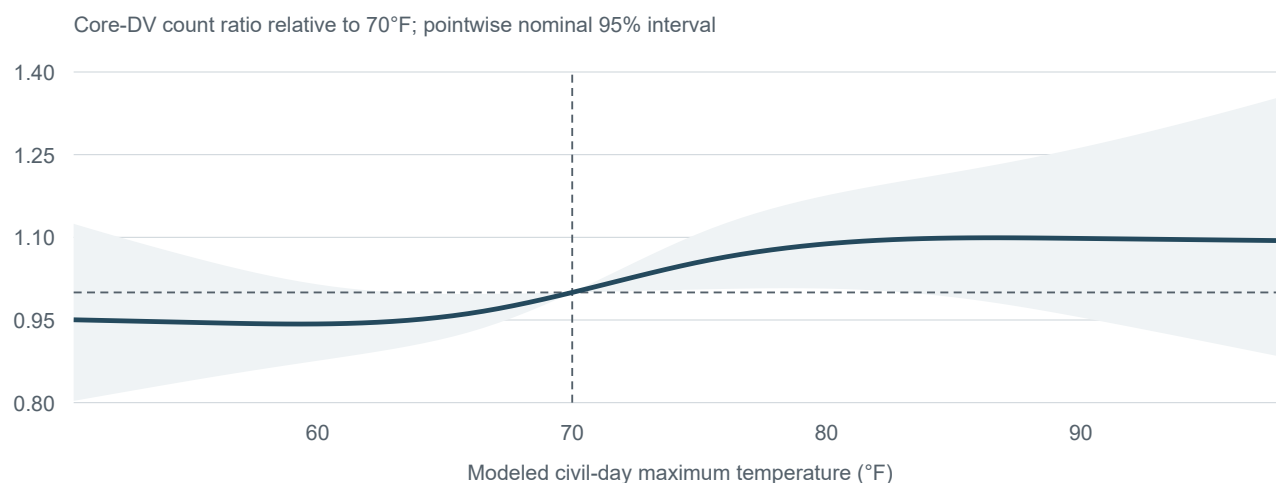


Figure 1. Restricted-cubic-spline core-DV count ratio relative to 70°F. The shaded region is a pointwise nominal 95% interval. Display is restricted to the pooled weather 1st–99th percentile range; no simultaneous inference or extrapolation beyond that range is claimed.

Spline hypothesis	Degrees of freedom	Raw <i>p</i>	Joint-family Holm <i>p</i>
Any temperature association	3	0.181	1.000
Departure from linearity	2	0.313	1.000

A nonsignificant curvature test does not validate linearity or prove absence of a threshold. The curve and its uncertainty should be interpreted over the observed range, with the same source and exposure limitations as the linear models.

Count-model diagnostic summaries

Model	Pearson dispersion	Observed zeros	Poisson fitted zeros	Daily residual lag 1
R0	1.031	93.6%	93.6%	−0.033
R5	0.969	93.5%	93.5%	−0.047
R6	0.969	93.5%	93.5%	−0.048

Seasonal core-DV fits required extended maximum likelihood: two ZIPs each had one positive outcome at a single annual phase. Certified separation removed 2,918 zero-count rows (2,893 in paired fits) and redundant seasonal columns; all positive outcomes remain. This amendment followed a failed fit. The mean model and 24-test family were retained. Diagnostics describe reduced fitted rows, not the full calendar. [21]

8. Station and covariance sensitivity analyses

For each retained ZIP, select the nearest of six previously acquired NOAA stations by geographic distance, breaking distance ties by station ID. Require distance ≤ 25 km and absolute station-versus-modeled-point elevation difference ≤ 200 m; do not substitute another station after this screen. Invalid or missing station maximum values exclude the same rows from both R7 and R8. [12,13,21]

Paired station analysis	Modeled maximum, R7	NOAA maximum, R8
Count ratio per 10°F (95% CI)	1.065 (0.979–1.159)	0.979 (0.912–1.051)
DV identifiers	3,809	3,809
Fitted ZIPs / ZIP-days	46 / 63,936	46 / 63,936
Raw p / joint-family Holm p	0.142 / 1.000	0.557 / 1.000

The identical fitted-row hash and outcome count are checked before reporting. NOAA station reporting-day conventions and spatial representation can differ from civil-day representative-point weather. This is an alternate-source sensitivity on a selected subset, not validation of personal exposure, a correction derived from station bias, or an independent holdout test. [12,13,21]

The original six-station same-location agreement comparison showed meaningful modeled-versus-observed differences, including daily-high mean absolute errors of approximately 2.6–5.9°F. Those historical checks motivate alternate-source sensitivity but do not establish the direction of bias in a small temperature association. No station-average offset was applied to the revised modeled temperatures. [21]

HAC lag sensitivity

Model	HAC lag, days	Count ratio (95% CI)	Raw p
R0	7	1.027 (0.988–1.069)	0.179
R0	14	1.027 (0.987–1.069)	0.187
R0	28	1.027 (0.985–1.071)	0.203
R5	7	1.048 (0.988–1.112)	0.116
R5	14	1.048 (0.986–1.115)	0.133
R5	28	1.048 (0.980–1.122)	0.169

HAC 14 / HAC 28 change covariance bandwidth, not fitted coefficients. They are diagnostic uncertainty comparisons outside the 24-test family; no multiplicity-adjusted claim is made for these additional intervals. Calendar gaps remain represented in the daily score sequence.

9. Unchanged descriptive temporal patterns

These descriptions use all core-DV source IDs with usable recorded timing in 2021–2024, before weather/location eligibility. They preserve the original summaries and do not shrink to the revised weather subset. Counts per weekday divide by the number of corresponding calendar dates, including zero-count days. [1,21]

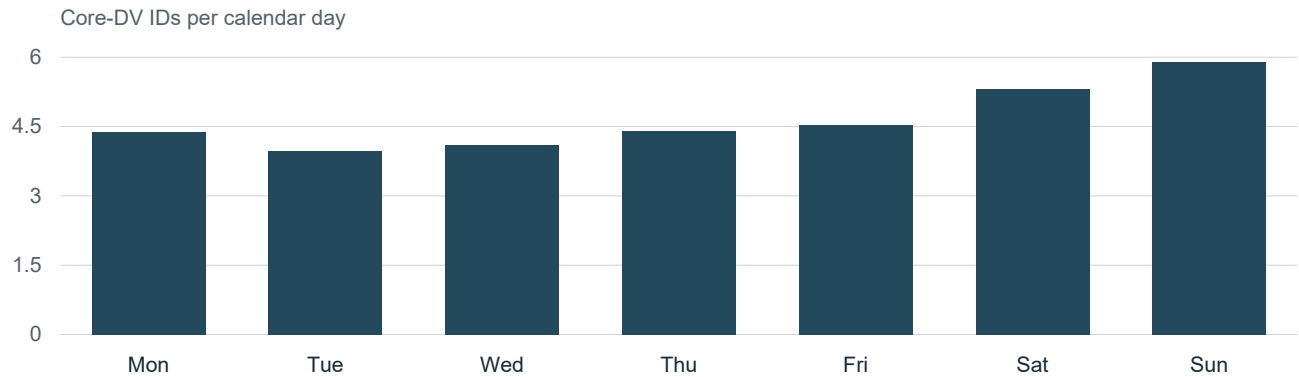


Figure 2. Recorded weekday, 2021–2024. Sunday averaged 5.89 IDs/day and Tuesday 3.98. Weekends averaged 5.61/day versus 4.28 on weekdays: an unadjusted 31.0% difference. Recorded dates are not verified offense dates.

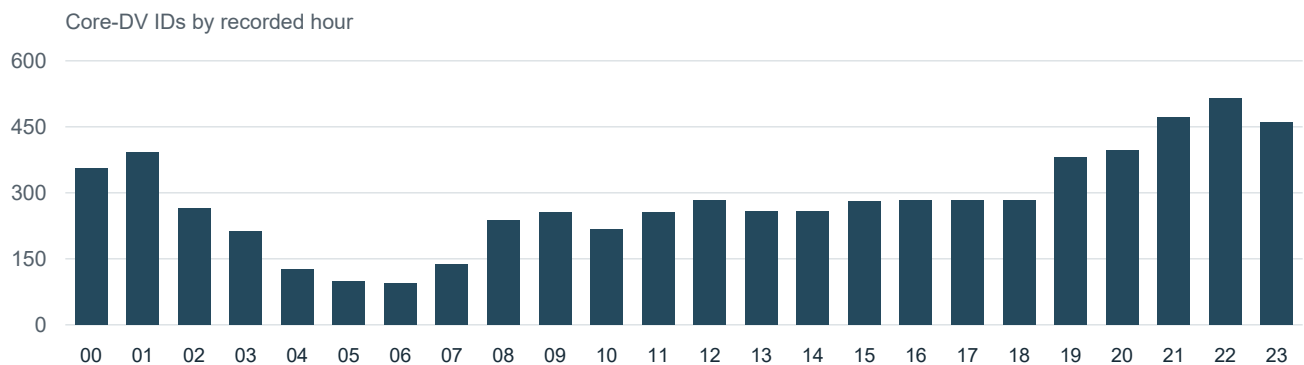


Figure 3. Recorded hour, 24-hour clock. The largest bin is 22:00–22:59 with 515 IDs. One core-DV ID has conflicting hours and is omitted from this figure. The display cannot establish when violence occurred.

Season and recorded location

July had the highest monthly count per calendar day, 5.39 core-DV IDs/day. A seasonal pattern may reflect heat, other seasonal influences or enforcement/reporting differences; it is not an independent causal test. Across all supplied years, Vista has 1,727 core-DV IDs, San Marcos 1,126 and Spring Valley 1,118. The administrative label SHERIFF has 829. These are recorded labels, not population rates or complete municipal totals.

Vista has 19,315 all-record groups and San Marcos 9,883. Core DV represents 8.9% and 11.4%, respectively. The larger Vista count does not demonstrate a larger DV proportion. The reason for the overall recording-volume difference remains unresolved; neither duplicate charges nor identified warrants alone explain its lead.

10. Interpretation, limitations and research context

What the revised comparisons address

Matched old/new weather separates some exposure changes from geographic selection. R4 addresses identified warrant co-charges; R2/R3 evaluate specified covariates; R6 examines shape; R7/R8 compare sources on matched rows; R9 examines elapsed-day duration. These changes cannot recover missing records, define unverified source fields, or distinguish reporting and enforcement processes from violence occurrence.

The original full-sample daily-high interval included no association. In the revised restricted sample, the civil-day reference interval includes no association. No DV revision test met the 0.05 threshold after the joint Holm correction. Statistical nonsignificance does not demonstrate equivalence or establish that heat has no relationship with DV. Conversely, a positive estimate or nominally narrow interval does not establish causation.

Remaining measurement and design limits

Completeness is unverified even during years with continuous supplied dates. Charge ascertainment may omit DV hidden in general charges. Identifiers can combine people or episodes; charges are allegations. Recorded clocks were assumed local but not agency-verified. The daily maximum may occur after a morning record, and the underlying violence may precede either that day or a prior-day exposure. The lag model improves ordering relative to recorded date only.

Availability-based weather restriction changes the target administrative population and favors high-volume ZIPs. Modeled grid exposure cannot represent indoor conditions, air conditioning or individual location. NOAA observations introduce their own location/day conventions. Static effects, federal holidays, humidity, precipitation and annual cycles do not eliminate changing enforcement, tourism, school calendars, service availability or other confounding. No causal adjustment set or population denominator was validated.

Relation to prior empirical research

Sanz-Barbero et al. studied Madrid time series of femicides, police reports and helpline calls and reported outcome/lag-dependent heat associations. Zhu et al. related annual mean temperature to prior-year self-reported IPV among 194,871 partnered women in India, Nepal and Pakistan. These differ in population, exposure duration and outcome from recorded-date Sheriff-labeled charge groups. This analysis is not a replication, systematic review or pooled comparison. [18,19]

Specific agency clarification still required

Obtain the original response/extraction specification; define Agency, both ID fields, date fields and location roles; identify multiple-arrestee structure and time zone; reconcile monthly totals/dispositions; explain 2025 coverage and source-system changes; request complete 2025–2026 records, offense timestamps, victim relationships and case-level DV flags. A draft clarification request was prepared but not sent. [21,26–29]

11. Provenance, reproducibility and test family

Official monthly activity reports count arrests with exclusions for detention/court/non-Sheriff areas, PC 849.5 detentions and reports still under review. Their unit and cutoff are not verified equivalents of the supplied charge/ID exports. For example, July 2023 reports 1,142 arrests versus 1,558 unambiguous-month source IDs in the export; a candidate geographic exclusion gives 1,263, and adding ADULT subtype gives 1,130. These differing filters are context checks, not a completeness denominator or an agency crosswalk. [26–28]

Complete joint revision family: 24 tests

Test	General raw p	General Holm p	DV raw p	DV Holm p
R00	0.053	1.000	0.180	1.000
R0	0.053	1.000	0.179	1.000
R1	0.248	1.000	0.044	0.964
R2	0.862	1.000	0.170	1.000
R3	0.023	0.533	0.144	1.000
R4	0.017	0.398	0.218	1.000
R5	0.399	1.000	0.116	1.000
R6 joint	0.748	1.000	0.181	1.000
R6 curvature	0.841	1.000	0.313	1.000
R7	0.960	1.000	0.142	1.000
R8	0.868	1.000	0.557	1.000
R9	0.053	1.000	0.179	1.000

The general outcome counts all eligible source-ID groups, not all crimes. Its tests are shown solely to disclose the shared multiplicity family used for DV interpretation. Original released tests and earlier post hoc diagnostics remain separate historical records.

Reproduction and verification record

The report reads `revision_results.json` and `panel_metadata.json`; `build_revised_report.py` performs no fitting. `prepare_panel.py` and `fit_revision.py` reconstruct aggregate inputs and all fits. Source hashes, exact headers, warrant flags, cellwise count reconciliations, weather provenance, station assignments, fitted-row hashes and convergence diagnostics are retained. Original source hashes are unchanged. The original 12-page DV PDF and fitted results remain archived. [21]

An independent implementation numerically checked all 22 fits, separation certificates and all 24 adjusted tests; separate reference refits agreed. This checks computation, not the agency records. Record-level flags contain private source IDs and are excluded from public outputs. Only aggregate analysis inputs/results should be published. Computational agreement tests assess numerical implementation; they are not independent human peer review, proof of source completeness, or validation of causal assumptions.

12. References and source attribution

External titles are clickable. Legal sources support the operational charge taxonomy; they do not adjudicate individual cases. Data and computational records are identified separately. Online sources accessed September 24, 2026.

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8. California Penal Code 13700. Law-enforcement domestic-violence relationship definition.
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10. Copernicus Climate Change Service / ECMWF. ERA5-Land hourly data. DOI: 10.24381/cds.e2161bac.
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 29. City of Oceanside Police Department. Staff report 26-1636, September 16, 2026. Describes regional NetRMS hosted by the Sheriff; does not define these exports.
 30. Python documentation: zoneinfo. IANA time-zone support and daylight-saving transitions.
 31. Copernicus Climate Change Service / ECMWF. ERA5 hourly single-level data. doi:10.24381/cds.adbb2d47. Separate precipitation source.
 32. California Penal Code, fugitive proceedings. Sections 1551 and 1551.1 distinguish warrant and warrantless procedures.
 33. California Penal Code, warrant of arrest. Section 827.1: citation release on a misdemeanor arrest warrant.
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